Verification Form #3 - PA3

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Section: 001

Language: C#

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① Rotate 90° CCW about (1,3)
② Reflect across the line through (0,2) w/ direction (1,1)
③ Scale by $S_x = 2$ $S_y = \frac{1}{2}$

① Step Matrices

write T(1,3); R(90°); T(-1,-3) then mult to get $M_1 = T(1,3)R(90°)T(-1,-3)$...

3×3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad T(t_x, t_y) = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \quad R(\theta) = \begin{bmatrix} \cos & -\sin & 0 \\ \sin & \cos & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S(s_x, s_y) = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\sin(90) = 1$
$\cos(90) = 0$

$$T(1,3) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad T(-1,-3) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \quad R(90°) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 3 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$M_1 = \begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Find the unit direction u, uu^T, $F = 2uu^T - I$ (2×2) then write F as a 3×3 mult to get

$M_2 = T(0,1)\,F\,T(0,-2)$

$u = \begin{bmatrix} \frac{dx}{\|d\|} \\ \frac{dy}{\|d\|} \end{bmatrix}$ $\quad d = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

direction = (1,1)
through = (0,2)

$u^T = \begin{bmatrix} \frac{dx}{\|d\|} & \frac{dy}{\|d\|} \end{bmatrix}$ $\quad u = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ $\quad u^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

$\|d\| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\sqrt{2} \cdot \sqrt{2} = \sqrt{4} = 2$

$uu^T = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 1/2 & 1/2 \end{bmatrix}$

$2uu^T = 2\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $\quad T(0,2) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

$2uu^T - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\quad T(0,-2) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$

make it 3×3 $= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ → R_3 from 3×3 I

$M_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} =$

$$\boxed{M_2 = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}}$$

Reflect (0,0) w/ M_2 then Confirm the answer is the Reflection of (0,0)

$$M_2 = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad P' = M_2 P = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$$

$3\times3 \quad 3\times1 \quad 3\times1$

$$P = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ but $\times 3$

$P' = (-2, 2)$

$q = (0,2)$
$d = (1,1)$

$$v = P - Q = (0,0) - (0,2) = (0,-2)$$

$$\text{Proj}_d(v) = \frac{v \cdot d}{d \cdot d} d = (-1,-1).$$

$$\frac{\begin{bmatrix} 0 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{0 + -2}{1+1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\frac{-2}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$H =$

$$q + \text{proj}_d(v) = (0,2) + (-1,-1) = (-1,1)$$

$$P' = 2H - P = 2(-1,1) - (0,0)$$
$$(-2,2) - (0,0) = (-2,2)$$

$P' = (-2,2)$

Both P' are equal! $P' = (-2,2)$

(3.) Check Step-by-Step

Vertex 3 = (2,5) $M_1 = \begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ $M_2 = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ $M_3 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$v = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 2 \\ 5 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $M = \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 3 \\ 0 & 0 & 1 \end{bmatrix}$

Result

$M_1 X = \begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} -5+4 \\ 2+2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ (-1, 4)

$M_2(M_1 X) = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 4-2 \\ -1+2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ (2, 1)

$M_3(M_2 M_1 X) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(2) \\ \frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{1}{2} \\ 1 \end{bmatrix}$ $(4, \frac{1}{2})$

M x

Compose Mx $\begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{1}{2}(5)+3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{1}{2} \\ 1 \end{bmatrix}$ $(4, \frac{1}{2})$

Do they agree? Yes! they agree $\begin{bmatrix} 4 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{1}{2} \end{bmatrix}$

(4.) Changing the order

Swap the first two steps Compute $M' = M_3 M_1 M_2$ and apply M' to the vertex 3.

$M_1 M_2 = \begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Vertex 3 = $\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$

$$M_3(M_1 M_2) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 4 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M'x = \begin{bmatrix} -2 & 0 & 4 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} -4+4 \\ \frac{1}{2}(5) \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5/2 \\ 1 \end{bmatrix} = (0, 5/2)$$

Is M' equal to M? No $\begin{bmatrix} -2 & 0 & 4 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \neq \begin{bmatrix} 2 & 0 & 0 \\ 0 & -\frac{1}{2} & 3 \\ 0 & 0 & 1 \end{bmatrix}$

IS M'x equal to Mx? No $(0, 5/2) \neq (4, 1/2)$

What is the distance between the two image of vector 3? $\underset{x_1\ y_1}{(0, \frac{5}{2})}$ $\underset{x_2\ y_2}{(4, \frac{1}{2})}$ $= \sqrt{20}$ from each other

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(4-0)^2 - (\tfrac{1}{2} - \tfrac{5}{2})^2}$$

$$= \sqrt{4^2 + (-\tfrac{4}{2})^2} = \sqrt{16 + (4)} = \sqrt{20}$$

$\quad\quad\quad

$$A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} d & -b \\ c & a \end{bmatrix} = \frac{1}{-1}\begin{bmatrix} -\frac{1}{2} & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -2 \end{bmatrix}$$

$$A^{-1}p = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ -6 \end{bmatrix} \qquad -A^{-1}p = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$\det(A) = -1 \quad A^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -2 \end{bmatrix} \quad A^{-1}p = \begin{bmatrix} 0 \\ -6 \end{bmatrix}$

$$M^{-1} = \begin{bmatrix} A^{-1} & -A^{-1}p \\ 0\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -2 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

check... $M^{-1}M = I$ then apply M^{-1} to the image to recover (2,5)

$$\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -2 & 6 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ - True}$$

$$Mx = \begin{bmatrix} 4 \\ 1/2 \\ 1 \end{bmatrix} \quad \text{apply } M^{-1}(Mx) = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -2 & 6 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(4) \\ -1+6 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

We get original vector! $\longrightarrow$ (2,5)

(6.) The Triangle Map

from V and V^{-1} find $\det(V)$, $\det(V^{-1})$, $A = V'V^{-1}$

$b_1 - Aa_1$ as a 3×3

$V = [a_2 - a_1 \quad a_3 - a_1]$

from points given.... $V = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

$V = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

$V' = \begin{bmatrix} 2 & 4 \\ -2 & 5 \end{bmatrix}$

$\det(V) = 6$

$V^{-1} = \begin{bmatrix} 1/2 & -1/6 \\ 0 & 1/3 \end{bmatrix}$

$\det(V^{-1}) = \frac{1}{6}$

$A = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$

$V' = [b_2 - b_1 \quad b_3 - b_4]$

$V' = \begin{bmatrix} 2 & 4 \\ -2 & 5 \end{bmatrix}$

$\det(V^{-1}) = \frac{1}{2}\left(\frac{1}{3}\right) - 0$

$\frac{1}{6}$

$\det(V) = ad - cb = 6$

$= 2(3) - (1)(0) =$

$= 6 - 0$

$= 6$

$V^{-1} = \frac{1}{\det(A)}\begin{bmatrix} d & -b \\ c & a \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}$

$V^{-1} = \begin{bmatrix} 1/2 & -1/6 \\ 0 & 1/3 \end{bmatrix}$

$A = V'V^{-1} = \begin{bmatrix} 2 & 4 \\ -2 & 5 \end{bmatrix}\begin{bmatrix} 1/2 & -1/6 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$

$P = b_1 - Aa_1$

$P = \begin{bmatrix} 5 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$P = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$a_1 = [1 \quad 1]$

$b_1 = [5 \quad 2]$

$A \quad a_1$

$\begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1 \\ -1+2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Form.... $\begin{bmatrix} a_{11} & a_{12} & P_1 \\ a_{21} & a_{22} & P_2 \\ 0 & 0 & 1 \end{bmatrix} = M_T$

$\begin{bmatrix} 1 & 1 & 3 \\ -1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Check... apply to a_2, a_3 to get b_2, b_3

$a_2 = (3,1)$

$b_2 = (7,0)$

$M_T a_2 = \begin{bmatrix} 1 & 1 & 3 \\ -1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3+1+3 \\ -3+2+1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \\ 1 \end{bmatrix} = b_2$

$a_3 = (2,4)$

$b_3 = (9,7)$

$M_T a_3 = \begin{bmatrix} 1 & 1 & 3 \\ -1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+4+3 \\ -2+8+1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \\ 1 \end{bmatrix} = b_3$

(7.) Compare

	Hand	Program	Match?
M_1	$\begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	yes
M_2	$\begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	yes
M_3	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	yes
M	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -\frac{1}{2} & 3 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -0.5 & 3 \\ 0 & 0 & 1 \end{bmatrix}$	yes
M_x	$\begin{bmatrix} 4 \\ \frac{1}{2} \\ 1 \end{bmatrix}$		
M'_x	$\begin{bmatrix} -2 & 0 & 4 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -2 & 0 & 4 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	yes
M^{-1}	$\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -2 & 6 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & -2 & 6 \\ 0 & 0 & 1 \end{bmatrix}$	yes

The methods in the code were off,

started w/ 100 not 10 Tx
010 01 Ty
TxTy1 001

when I change those everything matched.